



Distributed Systems and Internet technologies

COMMUNICATIONS

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COMMUNICATIONS

- We will use conventional communication tools / libraries from Java.
- Android does not provide any web services library. To use third-party providers.
- For local communications between the emulator and a server running on the localhost, the server will be accessible on IP: 10.0.2.2



COMMUNICATIONS

- Example of HTTP communication using the package **java.net**:

```
URL url = new URL("http://localhost:8080/ServiceExample/login.jsp");
HttpURLConnection ucon = (HttpURLConnection) url.openConnection();
ucon.setRequestMethod("POST");
ucon.setDoInput(true);
ucon.setDoOutput(true);
ucon.setRequestProperty("Content-Type", "application/json");
ucon.setRequestProperty("Accept", "application/json");
OutputStream os = ucon.getOutputStream();
...
ucon.connect();
InputStream in = ucon.getInputStream();
…
ucon.disconnect();
```

COMMUNICATIONS

- Transmission and reception of data is most commonly done in JSON format.
- The **gson** library from Google easily translates any Java object to its **JSON** content and the other way around. This way, we can send a plain text content to rebuild the Java object at destination.

EXERCISE

- For the RPC calls we will use the **ProgressDialog** class to indicate a communication in progress, and we will use an independent executing thread to program the connection to the remote server
- Analyze the coding of b_LoginActivity, intended for the remote call of the method login().
- Try and understand the use of Handler and **OperationPerformer**; and the way the auxiliary thread sends data to the UI thread.
- The alternative: the use of the **AsyncTask** class to implement the same behavior. This is what we will use to code the d_UsersListActivity.

EXERCISE

- Complete the code of the **b_LoginActivity** using OperationPerformer to log into the system.
- Complete the code of the **c_RegistrationActivity** using OperationPerformer to register a new user.